# Bharat Sirmal

Navi Mumbai | sirmalbharat99@gmail.com | LinkedIn | GitHub | Portfolio

## Education

**Pillai College of Engineering, Panvel** 2023 – 2027
*B.Tech in Computer Science – CGPA: 8.74 / 10.0*

**KMC Balkumari, Lalitpur, Nepal** 2020 – 2022
*Class 12 (Higher Secondary) – GPA: 3.14 / 4.0*

## Skills

**Languages:** Python, C/C++, JavaScript, TypeScript, SQL
**AI / Machine Learning:** PyTorch, CNNs, RAG, LLMs
**Backend / Frontend:** Flask, FastAPI, Node.js, Express.js, REST APIs, React, Next.js, HTML, CSS, Tailwind CSS
**Databases / Tools:** MongoDB, MySQL, SQLite, ChromaDB, Firebase, Git, GitHub, Postman, Google Cloud, Vercel, Cloudinary

## Projects

**Explainable Brain Tumor Classification |** *PyTorch · CNN · MRI/NIfTI · Grad-CAM* | GitHub
- Developed a 2-stage multimodal AI pipeline that screens 3 MRI views for tumor presence and classifies positive screens into 4 tumor types, incorporating 2 clinical inputs—age and sex.
- Integrated 2 explainability methods, Grad-CAM and Shapley attribution, with scan-specific interpretations.
- Built a responsive dashboard with input verification, result-specific guidance, and automated PDF reporting.

**Bharat AI – Personal AI Assistant (RAG) |** *Python · Flask · Gemini · ChromaDB · Next.js* | GitHub
- Developed an end-to-end Retrieval-Augmented Generation (RAG) assistant using a private knowledge base, semantic retrieval, and Gemini for grounded responses.
- Built an embedding pipeline with all-MiniLM-L6-v2 and ChromaDB, including similarity-based retrieval and relevance filtering.
- Integrated the AI backend with a conversational Next.js interface supporting streaming responses and voice interaction.

**JoJo Store – Full-Stack E-Commerce Platform |** *Next.js · TypeScript · Firebase · Stripe* | GitHub
- Developed a responsive e-commerce platform with product browsing, shopping cart, secure authentication, and Stripe payment integration.
- Built an admin dashboard for managing products, inventory, customers, orders, and delivery assignments.
- Integrated Firebase for authentication and data storage, Cloudinary for image uploads, and role-based access for customers, administrators, and delivery partners.
- Implemented role-based access for 3 user roles and validated core e-commerce workflows with 64 automated tests.

**YAOP Community Platform |** *Next.js · Express.js · MongoDB · Firebase* | GitHub
- Designed and managed a centralized MongoDB database, organizing structured data for events and volunteers with full accuracy and consistency across all records.
- Processed and validated donation entries with complete data integrity by implementing Firebase authentication and structured data workflows.
- Built an admin dashboard that streamlined data entry and record updates, reducing manual effort by ~40% and minimizing input errors.

## Internship

**Web Development Intern |** May 2026 – Jun 2026
*InAmigos Foundation | Remote*
- Assisted in developing and maintaining the organization's website using HTML, CSS, and JavaScript. Designed and implemented new features to enhance site functionality, and improved the layout and structure to make the website more visually attractive and easier to manage.
- Technology Used: HTML, CSS, JavaScript, React, Node.js, MongoDB

## Certifications

Google Cloud Data Analytics Certificate – Link

Google Cloud Career Launchpad (Foundations Track) – Link
Microsoft Power BI – The Complete Guide (Infosys Springboard) – Link
Python 3.4.3 Training — IIT Bombay (Spoken Tutorial), Oct 2024 – Link